Length Contraction and Relative Velocities

Worked answer key: question, given, diagram, steps with reasons, final answer

Conventions used throughout

• **Notation:** subscript s = measured by an observer at rest relative to the object or clock (proper length L_s, proper time Δt_s); subscript m = measured by an observer the object moves past. v = the object's speed relative to the observer, c = the speed of light, and γ (gamma) is short for $1/\sqrt{(1 - v^2/c^2)}$, which is always 1 or more.

• **Formulas:** $L_m = L_s/\gamma$ (lengths along the motion only), $\Delta t_m = \gamma \Delta t_s$, $p = \gamma mv$ (p = momentum, m = mass).

• **Relative velocities:** v_{XY} is the velocity of X relative to Y, and $v_{YX} = -v_{XY}$. X, Y and Z stand for any three objects; each question names its own objects with its own letters. When a label is a number, a comma separates the two labels: $v_{2,1}$ is ship 2 relative to ship 1. Along one line, $v_{XZ} = (v_{XY} + v_{YZ}) / (1 + v_{XY}v_{YZ}/c^2)$. Signs come from the stated + direction. At everyday speeds the denominator is about 1 and this becomes $v_{XZ} = v_{XY} + v_{YZ}$.

• **$c = 3.00 \times 10^8$ m/s.** Full precision is kept until the final answer.

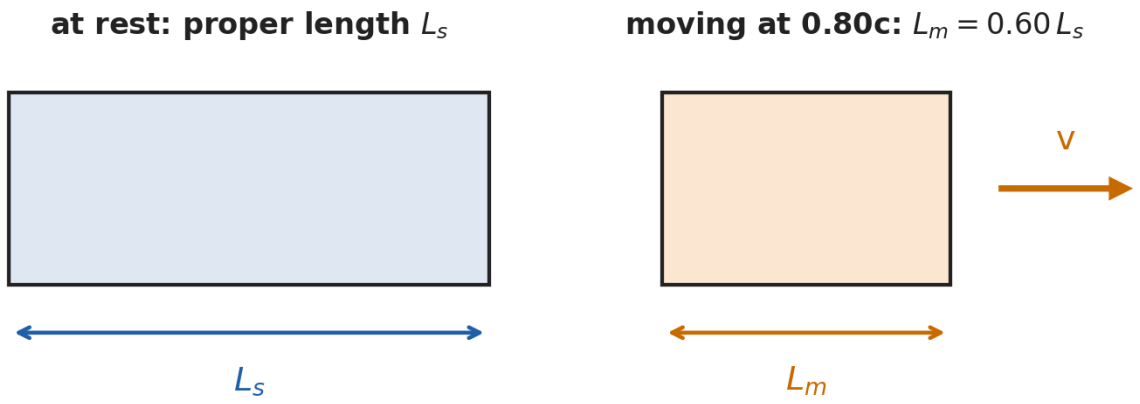
NOTES

Proper and contracted lengths

Length contraction applies only to lengths measured parallel to the direction of the velocity. Lengths measured perpendicular to the velocity are not affected.

To simplify problems involving relativistic speeds, physicists have assigned the symbol γ (gamma in the Greek alphabet) to the value of $1/\sqrt{(1-v^2/c^2)}$.

Diagram: the same box at rest and moving at 0.80c



Only the length along the motion shrinks; the height (perpendicular to v) is unchanged.

What the formulas say

γ (gamma) is a stretch factor that depends only on the speed. Because v is always less than c, v^2/c^2 is between 0 and 1, so the square root is less than 1 and γ is 1 or more. At everyday speeds γ is almost exactly 1 (no noticeable effect); near c it becomes large. Dividing a length by γ shortens it; multiplying a time by γ stretches it.

$$\gamma = \frac{1}{\sqrt{1-\frac{v^2}{c^2}}} \qquad L_m = \frac{L_s}{\gamma} = L_s\sqrt{1-\frac{v^2}{c^2}} \qquad \Delta t_m = \gamma\,\Delta t_s$$

Example: at v = 0.80c, $\sqrt{(1 - 0.64)} = 0.60$, so γ = 1.67 and a 10 m object is measured as 6.0 m long.

Moving objects are measured shorter along their motion only: $L_m = L_s/\gamma$, with $\gamma = 1/\sqrt{(1 - v^2/c^2)}$.

EXAMPLE

Two ships approaching

> If an observer on earth sees two space ships approaching each other and she measures the speed of ship 1 wrt (with respect to) herself to be 0.8c to the right and ship 2 to be 0.7c to the left. An observer in ship 1 also sees the approach of ship 2. What velocity does he see ship 2 moving at wrt his ship 1? Calculate the non-relativistic result as a multiple of c.

Picture the situation



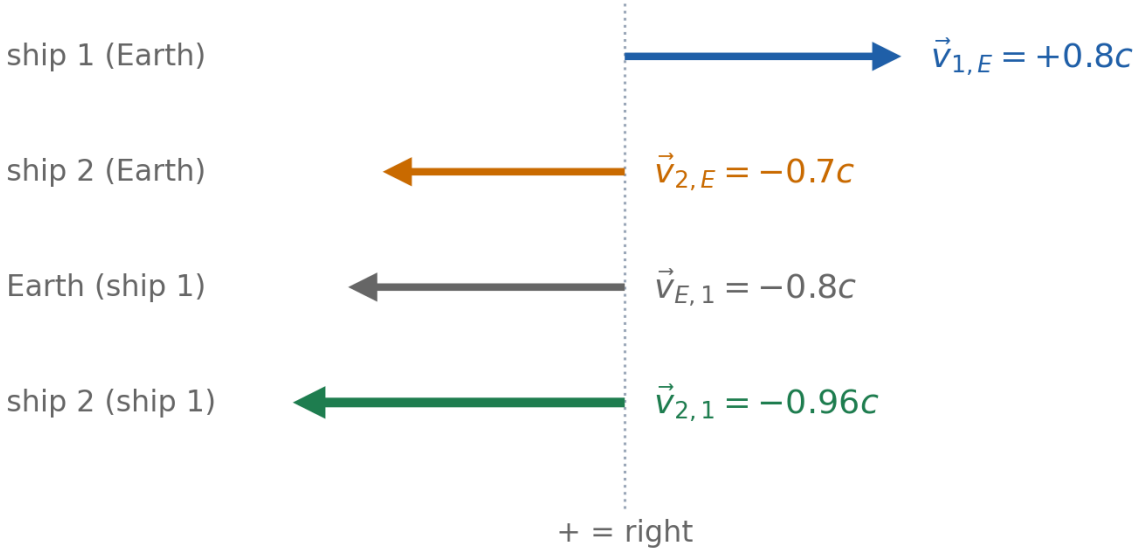
Two ships fly toward each other. Earth sees ship 1 going right at 0.8c and ship 2 going left at 0.7c. We want the speed of ship 2 as measured from ship 1.

GIVEN

$v_{1,E} = +0.8c$ $v_{2,E} = -0.7c$ $v_{2,1} = ?$

Sign convention: right is positive. 1 = ship 1, 2 = ship 2, E = Earth. Earth relative to ship 1 is $v_{E,1} = -0.8c$.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Non-relativistic (classical) result

The question asks for the everyday (classical) answer first. We want ship 2 relative to ship 1. We know ship 2 relative to Earth, and Earth relative to ship 1 is the reverse of ship 1 relative to Earth, so $v_{E,1} = -0.8c$. Chain them: ship 2 → Earth → ship 1, and add.

Formula (everyday speeds): $v_{XZ} = v_{XY} + v_{YZ}$

$$v_{2,1} = v_{2,E} + v_{E,1} = (-0.7c) + (-0.8c) = -1.5c$$

This says 1.5c to the left, faster than light, which is impossible. Classical addition fails at these speeds.

Step 2 – Relativistic result

Here every speed is close to c, so ordinary adding and subtracting of velocities gives wrong answers. We use the relativistic formula from the conventions box instead. It works exactly like the ordinary chain rule for relative velocities (X relative to Y plus Y relative to Z gives X relative to Z), except that we divide by 1 + (product of the two velocities)/c². Use the same two velocities as in Step 1.

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\,v_{YZ}}{c^2}}$

$$v_{2,1} = \frac{-0.7c + (-0.8c)}{1 + \dfrac{(-0.7c)(-0.8c)}{c^2}}$$

Top: $-0.7c - 0.8c = -1.5c$

Bottom: $1 + (-0.7)(-0.8) = 1 + 0.56 = 1.56$

$$v_{2,1} = \frac{-1.5c}{1.56} = -0.96c$$

Non-relativistic: 1.5c [left]

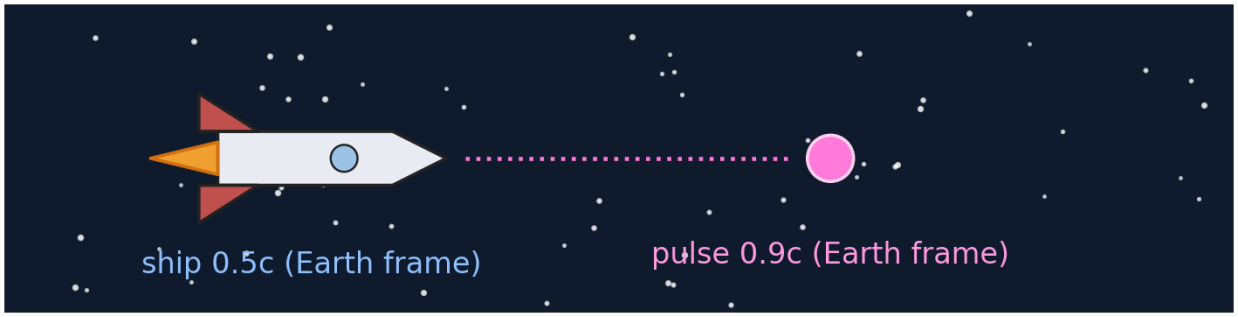
Relativistic (what he actually measures): 0.96c [left]

EXAMPLE

Plasma pulse as seen from the ship

Observer on earth sees a ship travelling at 0.5c [fwd] wrt the ground. A plasma cannon fires a pulse out the front of the ship so that the ground observer sees the pulse travel at 0.9c [fwd] wrt the ground. How fast do the observers in the ship see the pulse travel wrt the ship?

Picture the situation



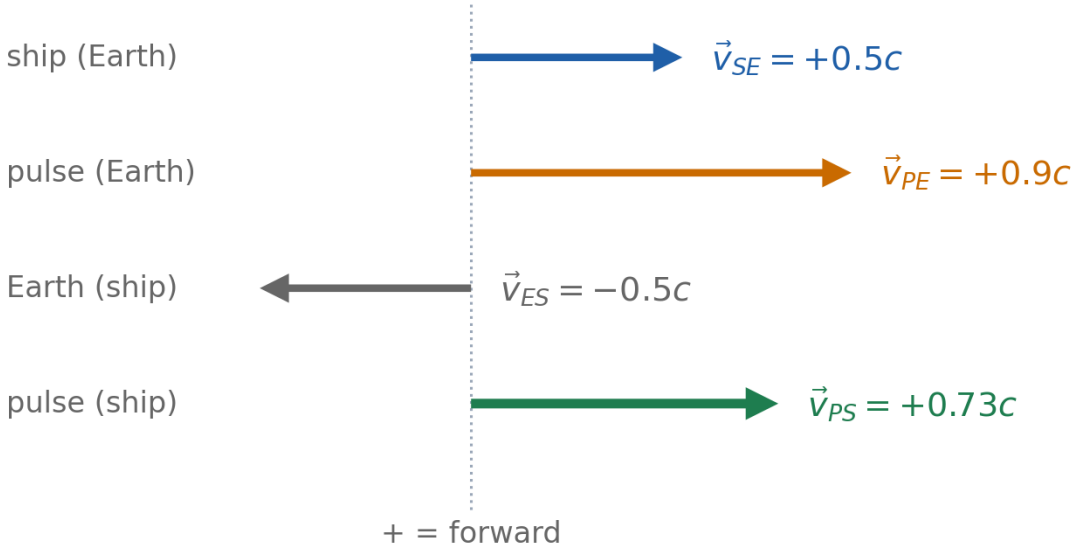
A ship fires a plasma pulse forward. Earth measures both speeds; we want the pulse's speed as measured by the crew.

GIVEN

$v_{SE} = +0.5c$ $v_{PE} = +0.9c$ $v_{PS} = ?$

Sign convention: forward is positive. S = ship, P = pulse, E = Earth (ground). $v_{ES} = -0.5c$.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Relativistic addition

These speeds are a large fraction of c, so we use the relativistic formula from the conventions box (same method as the Two-ships example). We want the pulse relative to the ship. Chain pulse → Earth → ship, using $v_{ES} = -0.5c$ (from the ship, Earth moves backward).

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\,v_{YZ}}{c^2}}$

$$v_{PS} = \frac{0.9c + (-0.5c)}{1 + \dfrac{(0.9c)(-0.5c)}{c^2}}$$

Top: $0.9c - 0.5c = 0.4c$

Bottom: $1 + (0.9)(-0.5) = 1 - 0.45 = 0.55$

$$v_{PS} = \frac{0.4c}{0.55} = 0.727c$$

0.73c [forward] relative to the ship

Classically the answer would be 0.9c − 0.5c = 0.4c. At these speeds velocities do not simply subtract; the relativistic formula gives the larger value, 0.73c.

QUESTION 1

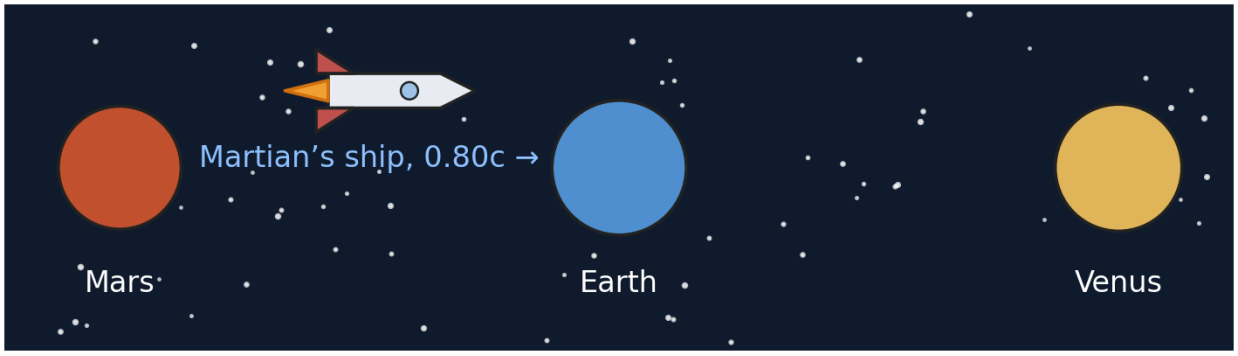
Mars-to-Venus trip

A Martian leaves Mars in a spaceship that is heading to Venus. On the way, the spaceship passes earth with a speed v=0.80c relative to it. Assume that the three planets do not move relative to each other during the trip. The distance between Mars and Venus is 1.20 x 10^{11} m, as measured by a person on earth.

(a) What does the Martian measure for the distance between Mars and Venus?

(b) What is the time of the trip (in seconds) as measured by the Martian?

Picture the situation

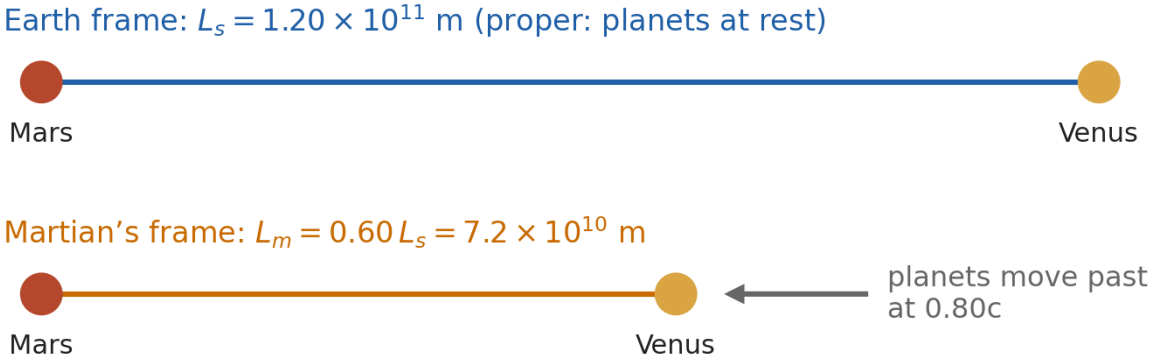


The planets are treated as sitting still in a line. A Martian's ship flies from Mars to Venus at 0.80c, passing Earth on the way.

GIVEN

$L_s = 1.20 \times 10^{11}$ m $\quad v = 0.80c \quad L_m = ? \quad \Delta t_s = ?$

Diagram: the distance in each frame



(a) Contracted distance

Who measures the proper length? The planets don't move relative to each other or to Earth, so the person on Earth is at rest relative to the Mars–Venus distance: that is L_s. The Martian flies past along that distance at 0.80c, so he measures it shorter. First work out the shrink factor, then multiply.

Formula: $L_m = L_s\sqrt{1-\frac{v^2}{c^2}}$

With $v = 0.80c$: $\frac{v^2}{c^2} = \frac{(0.80c)^2}{c^2} = 0.64$

$\sqrt{1-0.64} = \sqrt{0.36} = 0.60$

$L_m = (1.20\times10^{11}\text{ m})(0.60) = 7.20\times10^{10}\text{ m}$

(b) Trip time for the Martian

From the Martian's point of view he sits still and the planets slide past him at 0.80c, with Venus only 7.20×10^{10} m away (his shorter distance from part (a)). Time = distance ÷ speed. He is at both events (leaving Mars, reaching Venus), so this is the proper time on his clock.

Formula: $v = \frac{d}{\Delta t}$

Multiply both sides by Δt: $v\,\Delta t = d$

Divide both sides by v: $\Delta t = \frac{d}{v}$ (here $d = L_m$)

$\Delta t_s = \frac{7.20\times10^{10}\text{ m}}{(0.80)(3.00\times10^{8}\text{ m/s})} = \frac{7.20\times10^{10}}{2.40\times10^{8}} = 300\text{ s}$

(a) 7.2×10^{10} m

(b) 3.0×10^{2} s (5.0 min)

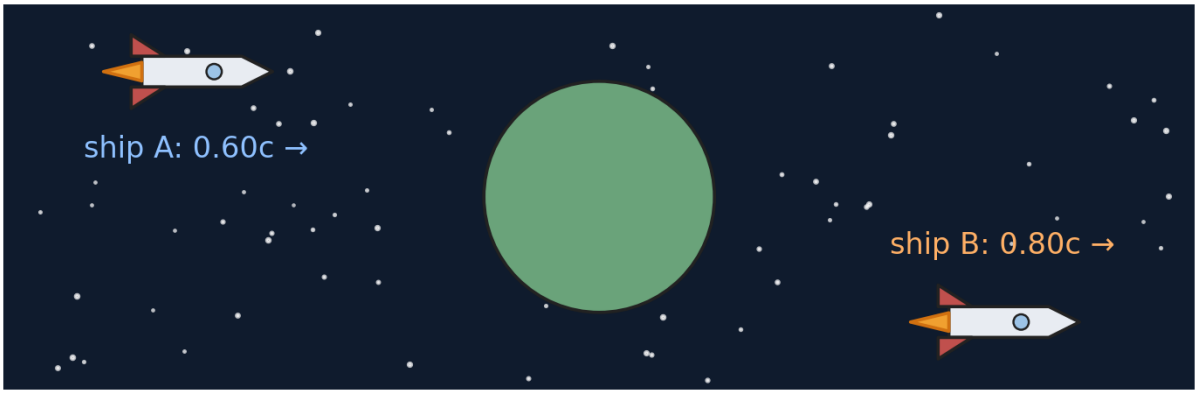
Check: Earth measures $1.20\times10^{11} \div (2.40\times10^{8}) = 500$ s. Time dilation gives the Martian's clock $500\times0.60 = 300$ s, the same answer by a different route.

QUESTION 2

Planet diameter seen by two ships

Two spaceships A and B are exploring a new planet. Relative to this planet, spaceship A has a speed of 0.60c, and spaceship B has a speed of 0.80c. What is the ratio D_A/D_B of the values for the planet's diameter that each spaceship measures in a direction that is parallel to its motion?

Picture the situation



Two ships fly past the same planet at different speeds. Each measures the planet's width along its own direction of motion.

GIVEN

$v_A = 0.60c$ $v_B = 0.80c$ $D_A/D_B = ?$

Step 1 – Each ship's measurement

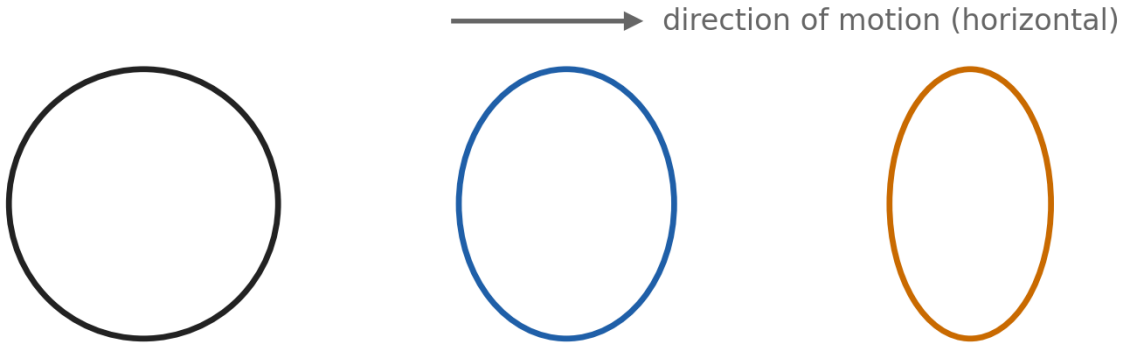
We don't know the planet's actual diameter, so call it D (its proper length, measured at rest). Each ship measures that diameter shrunk by its own factor $\sqrt{(1 - v^2/c^2)}$. The faster ship shrinks it more. In the formula, L_s is D, L_m is what the ship measures (D_A or D_B), and v is that ship's speed (v_A or v_B, measured relative to the planet).

Formula: $L_m = L_s\sqrt{1 - \frac{v^2}{c^2}}$

$$D_A = D\sqrt{1 - 0.60^2} = D\sqrt{1 - 0.36} = D\sqrt{0.64} = 0.80\,D$$

$$D_B = D\sqrt{1 - 0.80^2} = D\sqrt{1 - 0.64} = D\sqrt{0.36} = 0.60\,D$$

Diagram: the planet as measured by each ship



Step 2 – The ratio

Divide one measurement by the other. D appears on the top and bottom, so it cancels. That is why we never needed the planet's real size.

$$\frac{D_A}{D_B} = \frac{0.80\,D}{0.60\,D} = \frac{4}{3} = 1.33$$

D_A/D_B = 1.3 (exactly 4/3): the slower ship measures the larger diameter.

QUESTION 3

When momentum triples

At what speed is the magnitude of the relativistic momentum of a particle three times the magnitude of the nonrelativistic momentum?

Picture the situation



A particle speeds up. At everyday speeds p = mv works, but near the speed of light the true momentum grows much faster.

Step 1 – Compare the two momenta

Relativistic momentum is $p_{rel} = \gamma mv$, and the everyday (non-relativistic, or classical) momentum is $p_{classical} = mv$. The question says the first is three times the second. Divide one by the other: m and v cancel, leaving just γ.

Formulas: $p_{rel} = \gamma mv \qquad p_{classical} = mv$

$$\frac{p_{rel}}{p_{classical}} = \frac{\gamma mv}{mv} = 3 \Rightarrow \gamma = 3 \quad (m \text{ and } v \text{ cancel})$$

Step 2 – Solve for v

Now find the speed that makes γ = 3. Write out γ's formula, flip both sides and square to get rid of the square root, then solve for v^2/c^2. Take the positive square root because speed is positive.

Formula: $\gamma = \dfrac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$

$$\frac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}} = 3$$

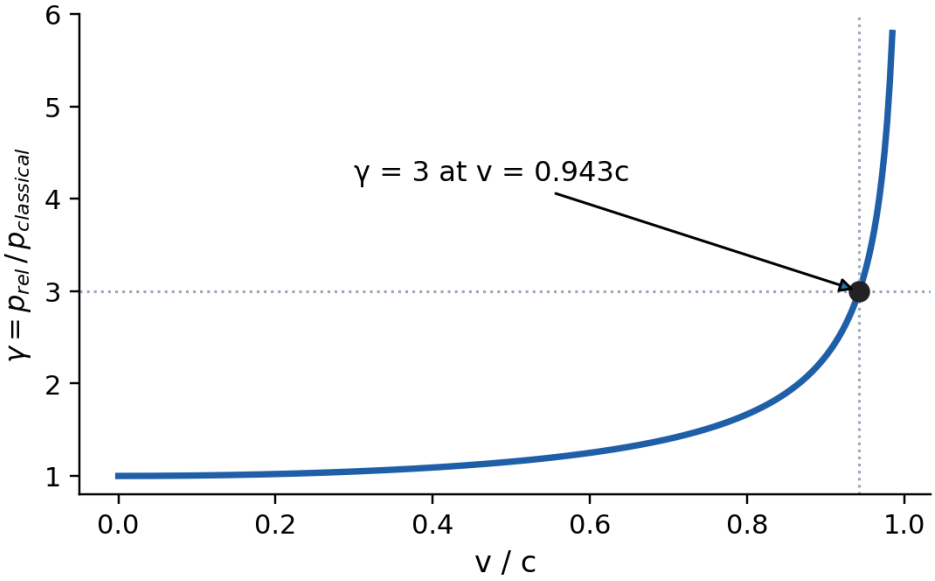
Flip both sides: $\sqrt{1 - \dfrac{v^2}{c^2}} = \dfrac{1}{3}$

Square both sides: $1 - \dfrac{v^2}{c^2} = \dfrac{1}{9}$

Rearrange: $\dfrac{v^2}{c^2} = 1 - \dfrac{1}{9} = \dfrac{8}{9}$

Square root, then multiply by c: $v = \sqrt{\dfrac{8}{9}}\, c = 0.943c$

Graph: γ (the momentum ratio) against v/c



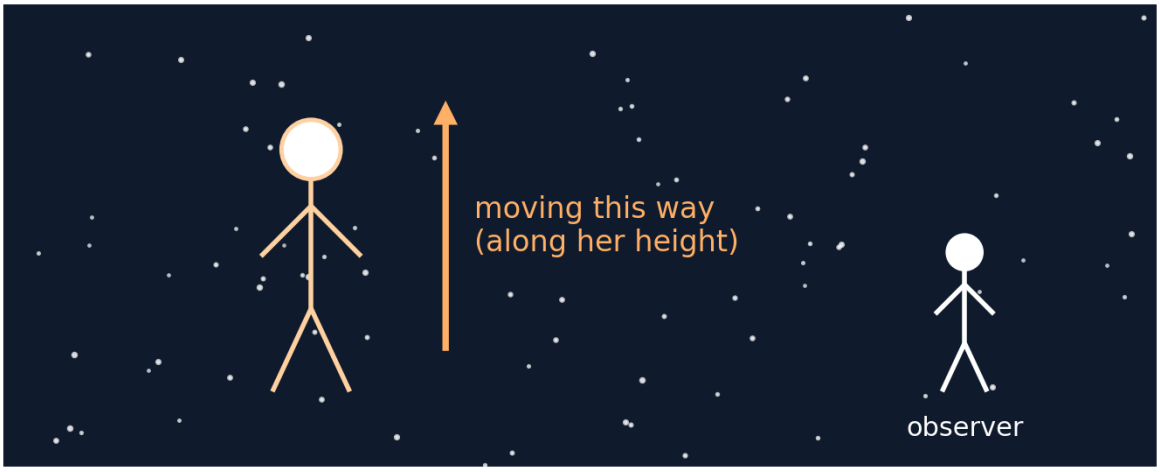
v = 0.943c (2.83 × 10^8 m/s)

QUESTION 4

Height measured by a moving observer

A woman is 1.6 m tall and has a mass of 55 kg. She moves past an observer with the direction of the motion parallel to her height. The observer measures her relativistic momentum to have a magnitude of 2.0 x 10^{10} kg m/s. What does the observer measure for her height?

Picture the situation



The woman zooms past the observer head-first, so her height lines up with her direction of motion.

GIVEN

$L_s = 1.6$ m $\qquad m = 55$ kg

$p = 2.0 \times 10^{10}$ kg m/s $\qquad L_m = ?$

Step 1 – Find her speed from the momentum

To find her contracted height we need γ, and for that we need her speed. The momentum tells us the speed: p = γmv = mv/√(1 − v²/c²). Multiply both sides by the square root, square both sides to remove the root, expand the bracket, collect the v² terms (factor out v²), and solve for v.

Formula: $p = \dfrac{mv}{\sqrt{1 - \dfrac{v^2}{c^2}}}$

Multiply both sides by the square root: $p\sqrt{1 - \dfrac{v^2}{c^2}} = mv$

Square both sides: $p^2\left(1 - \dfrac{v^2}{c^2}\right) = m^2v^2$

Expand the bracket: $p^2 - \dfrac{p^2v^2}{c^2} = m^2v^2$

Add $\dfrac{p^2v^2}{c^2}$ to both sides: $p^2 = m^2v^2 + \dfrac{p^2v^2}{c^2} = v^2\left(m^2 + \dfrac{p^2}{c^2}\right)$

Divide, then square root: $v = \dfrac{p}{\sqrt{m^2 + \dfrac{p^2}{c^2}}}$

$\dfrac{p}{c} = \dfrac{2.0 \times 10^{10}}{3.00 \times 10^8} = 66.67 \qquad v = \dfrac{2.0 \times 10^{10}}{\sqrt{55^2 + 66.67^2}} = \dfrac{2.0 \times 10^{10}}{86.43}$

$v = 2.31 \times 10^8$ m/s $= 0.771c$

Step 2 – Gamma

Put her speed (as a fraction of c) into γ's formula.

Formula: $\gamma = \dfrac{1}{\sqrt{1 - \dfrac{v^2}{c^2}}}$

$$\gamma = \frac{1}{\sqrt{1 - 0.7714^2}} = \frac{1}{\sqrt{1 - 0.5950}} = \frac{1}{\sqrt{0.4050}} = \frac{1}{0.6364} = 1.571$$

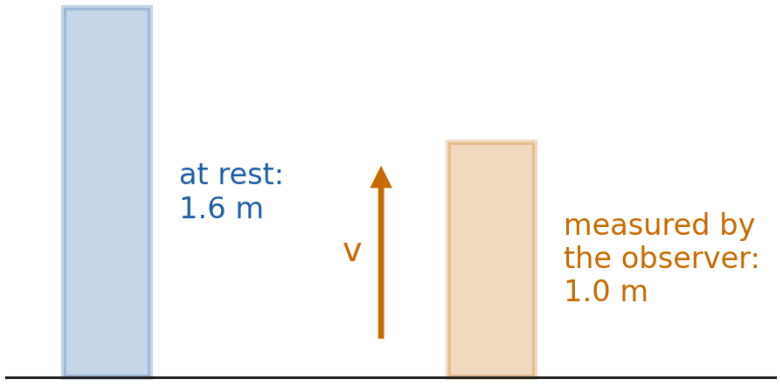
Step 3 – Contracted height

Her height is along her motion, so it contracts. 1.6 m is her proper height (she measures it at rest relative to herself). Divide by γ.

Formula: $L_m = \dfrac{L_s}{\gamma}$

$$L_m = \frac{1.6 \text{ m}}{1.571} = 1.02 \text{ m}$$

Diagram: her height at rest and as measured



The observer measures her height as 1.0 m.

Check: p/(mc) = 1.21 should equal γv/c = 1.571 × 0.7714 = 1.21. It does.

QUESTION 5

Oncoming truck with c = 65 m/s

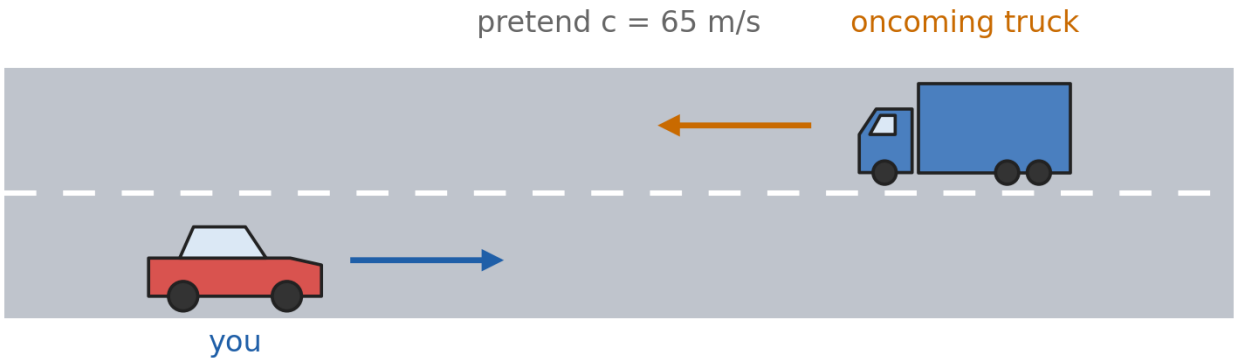
You are driving down a two-lane country road, and a truck in the opposite lane is traveling toward you. Suppose that the speed of light in a vacuum is c = 65 m/s. Determine the speed of the truck relative to you when

(a) your speed is 25 m/s and the truck's speed is 35 m/s and

(b) your speed is 5.0 m/s and the truck's speed is 55 m/s.

The speeds given in parts (a) and (b) are relative to the ground.

Picture the situation



An imaginary world where light is slow (65 m/s), so relativity matters at highway speeds. You and a truck drive toward each other.

GIVEN

$c = 65$ m/s

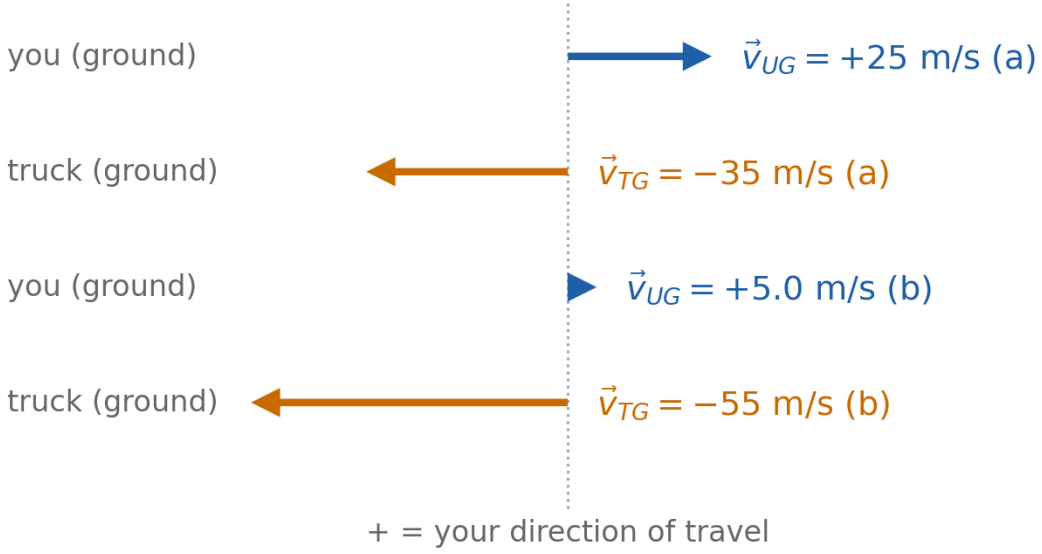
$v_{TU} = ?$

(a) $v_{UG} = +25$ m/s, $v_{TG} = -35$ m/s

(b) $v_{UG} = +5.0$ m/s, $v_{TG} = -55$ m/s

Sign convention: your direction of travel is positive. U = you, T = truck, G = ground. The truck comes toward you, so v_{TG} is negative, and $v_{GU} = -v_{UG}$.

Diagram: the ground velocities, one per row (tails lined up)



(a) You 25 m/s, truck 35 m/s

This is the same relativistic formula, but with c = 65 m/s instead of 3.00×10^8 m/s. We want the truck relative to you. Chain truck → ground → you. The ground relative to you is the reverse of your velocity, so $v_{GU} = -25$ m/s.

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\, v_{YZ}}{c^2}}$

$$v_{TU} = \frac{-35 + (-25)}{1 + \dfrac{(-35)(-25)}{65^2}}$$

Top: $-35 - 25 = -60$

Bottom: $1 + \dfrac{(-35)(-25)}{65^2} = 1 + \dfrac{875}{4225} = 1 + 0.2071 = 1.207$

$$v_{TU} = \frac{-60}{1.207} = -49.7 \text{ m/s}$$

(b) You 5.0 m/s, truck 55 m/s

Same steps with the new speeds: $v_{TG} = -55$ m/s and $v_{GU} = -5.0$ m/s.

$$v_{TU} = \frac{-55 + (-5.0)}{1 + \dfrac{(-55)(-5.0)}{65^2}}$$

Top: $-55 - 5.0 = -60$

Bottom: $1 + \dfrac{(-55)(-5.0)}{65^2} = 1 + \dfrac{275}{4225} = 1 + 0.0651 = 1.065$

$$v_{TU} = \frac{-60}{1.065} = -56.3 \text{ m/s}$$

(a) 5.0×10^{1} m/s toward you

(b) 56 m/s toward you

Classically both answers would be 60 m/s. With c this small, both relativistic answers are less than 60 m/s, and they are different from each other even though the speeds add to 60 m/s in both parts.

QUESTION 6

Exploration vehicle and spacecraft

A spacecraft approaching the earth launches an exploration vehicle. After the launch, an observer on earth sees the spacecraft approaching at a speed of 0.50c and the exploration vehicle approaching at a speed of 0.70c. What is the speed of the exploration vehicle relative to the spaceship?

Picture the situation



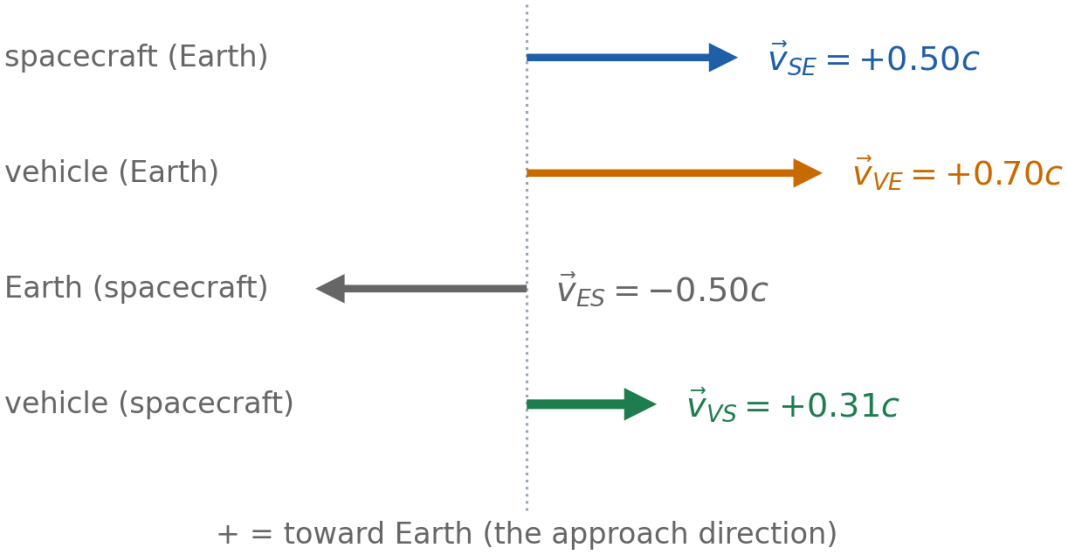
A spacecraft heading toward Earth sends a small vehicle out ahead of it. Earth measures both speeds.

GIVEN

$v_{SE} = +0.50c$ $v_{VE} = +0.70c$ $v_{VS} = ?$

Sign convention: toward Earth (the approach direction) is positive. S = spacecraft, V = vehicle, E = Earth. $v_{ES} = -0.50c$.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Relativistic addition

These speeds are a large fraction of c, so we use the relativistic formula from the conventions box (same method as the Two-ships example). We want the vehicle relative to the spacecraft. Chain vehicle → Earth → spacecraft, using $v_{ES} = -0.50c$.

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\,v_{YZ}}{c^2}}$

$$v_{VS} = \frac{0.70c + (-0.50c)}{1 + \dfrac{(0.70c)(-0.50c)}{c^2}}$$

Top: $0.70c - 0.50c = 0.20c$

Bottom: $1 + (0.70)(-0.50) = 1 - 0.35 = 0.65$

$$v_{VS} = \frac{0.20c}{0.65} = 0.308c$$

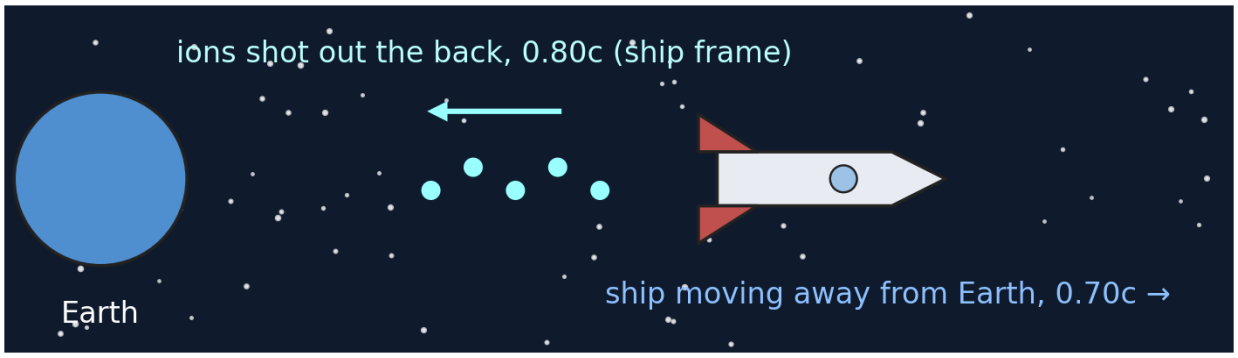
0.31c (toward Earth, ahead of the spacecraft)

QUESTION 7

Ions ejected backward

Spaceships of the future may be powered by ion-propulsion engines in which ions are ejected from the back of the ship to drive it forward. In one such engine the ions are to be ejected with a speed of 0.80c relative to the spaceship. The spaceship is traveling away from the earth at a speed of 0.70c relative to the earth. What is the velocity of the ions relative to the earth? Assume that the direction in which the spaceship is traveling is the positive direction, and be sure to assign the correct plus or minus signs to the velocities.

Picture the situation



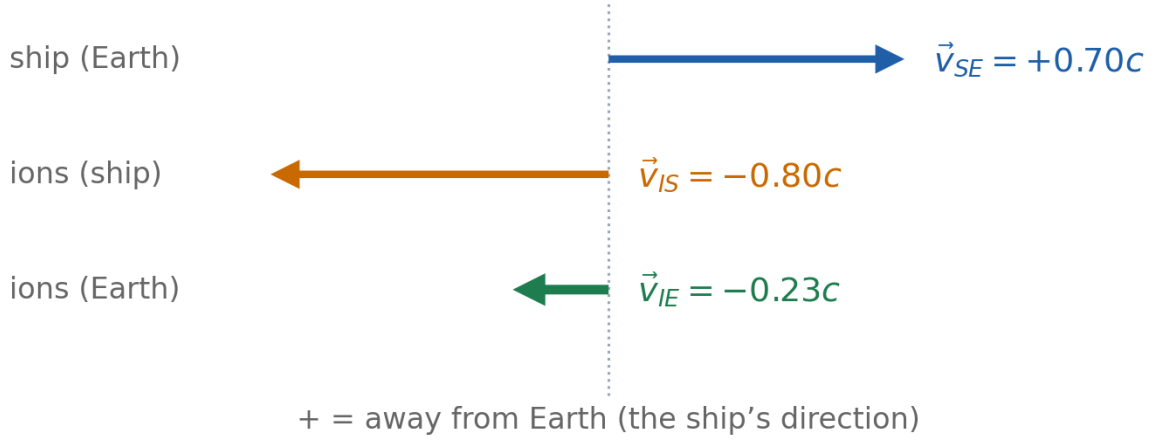
The ship flies away from Earth by firing ions out its back. The ions shoot backward relative to the ship.

GIVEN

$v_{SE} = +0.70c$ $v_{IS} = -0.80c$ $v_{IE} = ?$

Sign convention: the ship's direction (away from Earth) is positive, as the question says. I = ions, S = ship, E = Earth. The ions leave the back, so v_{IS} is negative.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Relativistic addition

These speeds are a large fraction of c, so we use the relativistic formula from the conventions box (same method as the Two-ships example). We want the ions relative to Earth. We know the ions relative to the ship (−0.80c, backward) and the ship relative to Earth (+0.70c). Chain ions → ship → Earth.

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\, v_{YZ}}{c^2}}$

$$v_{IE} = \frac{-0.80c + (0.70c)}{1 + \dfrac{(-0.80c)(0.70c)}{c^2}}$$

Top: $-0.80c + 0.70c = -0.10c$

Bottom: $1 + (-0.80)(0.70) = 1 - 0.56 = 0.44$

$$v_{IE} = \frac{-0.10c}{0.44} = -0.227c$$

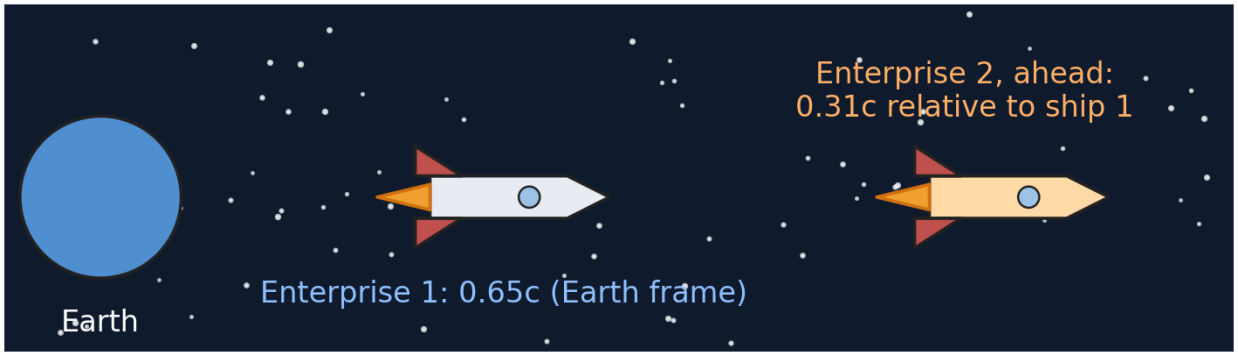
v_{IE} = −0.23c: the ions move toward Earth at 0.23c.

QUESTION 8

Two Enterprise ships

The spaceship Enterprise 1 is moving directly away from earth at a velocity that an earth-based observer measures to be 0.65c. A sister ship, Enterprise 2, is ahead of Enterprise 1and is also moving directly away from earth along the same line. The velocity of Enterprise 2 relative to Enterprise 1 is 0.31c. What is the velocity of Enterprise 2, as measured by the earth-based observer?

Picture the situation



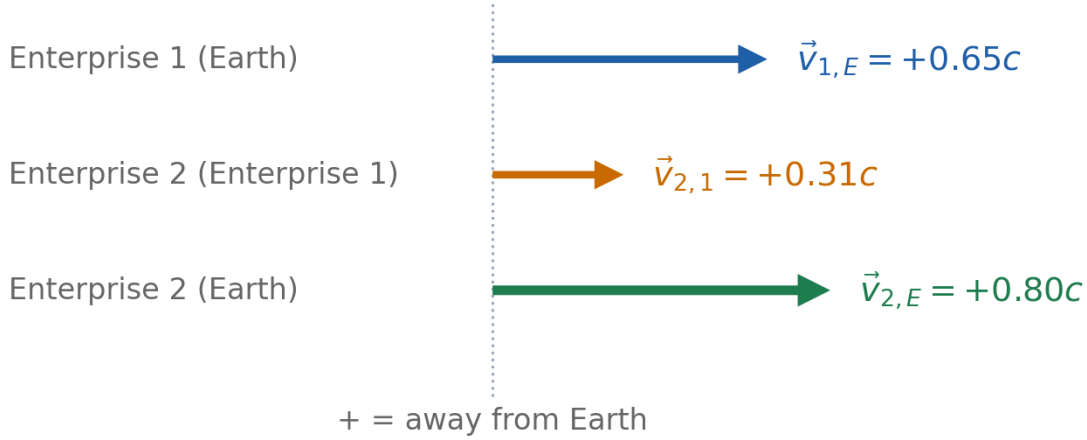
Two ships fly away from Earth in a line. Earth measures the first one; the second one's speed is given relative to the first.

GIVEN

$v_{1,E} = +0.65c$ $v_{2,1} = +0.31c$ $v_{2,E} = ?$

Sign convention: away from Earth is positive. 1 and 2 = the two ships, E = Earth.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Relativistic addition

These speeds are a large fraction of c, so we use the relativistic formula from the conventions box (same method as the Two-ships example). We want ship 2 relative to Earth. Chain ship 2 → ship 1 → Earth. Both velocities point away from Earth, so both are positive.

Formula: $v_{XZ} = \dfrac{v_{XY} + v_{YZ}}{1 + \dfrac{v_{XY}\, v_{YZ}}{c^2}}$

$$v_{2,E} = \frac{0.31c + (0.65c)}{1 + \dfrac{(0.31c)(0.65c)}{c^2}}$$

Top: $0.31c + 0.65c = 0.96c$

Bottom: $1 + (0.31)(0.65) = 1 + 0.2015 = 1.2015$

$$v_{2,E} = \frac{0.96c}{1.2015} = 0.799c$$

0.80c, directly away from Earth

Check: classically it would be 0.96c; the relativistic result is smaller and still below c.